

TOWARDS GLOBAL GEOMETRY PRESERVATION IN TEXT EMBEDDING DISTILLATION

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ABSTRACT

A text embedding model encodes relational knowledge in the global geometry it induces over a corpus. Relational knowledge distillation (KD) offers a natural way to transfer this knowledge, but existing objectives typically construct relations only among examples that share a mini-batch. Random batch co-occurrence rather than teacher relevance therefore selects the supervision support, leaving much of the corpus-level geometry outside the realized objective. We show that mini-batch exposure is independent of teacher relevance and bound global teacher-weighted distortion by the error on selected local relations plus the teacher mass they leave uncovered. Motivated by this view, we introduce **RIPPLE**, which approximates global geometry through teacher-defined local geometries. A cached mutual-neighbor graph defines immediate neighborhoods, diffusion expands them across scales, and sparse candidate pooling keeps training tractable. **RIPPLE** requires neither task labels nor online teacher inference and adds no inference-time cost. Across nine datasets and three teacher–student pairs, **RIPPLE** obtains the best mean score among the evaluated compact students in all three settings, with the strongest and most consistent gains on semantic textual similarity.

1 INTRODUCTION

Text embedding models map examples into a space whose global arrangement supports similarity search, clustering, and transfer across tasks. A teacher therefore contains knowledge not only in individual vectors, but also in the relations it induces among corpus examples. Distilling these relations should preserve the geometry of the teacher embedding space over the corpus, rather than merely reproduce isolated representations (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Kim et al., 2023; 2024).

Directly aligning all corpus relations is prohibitively expensive, so relational KD commonly constructs pair matrices or conditional distributions from the current mini-batch (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Qian et al., 2022; Gou et al., 2025). Reusing encoded examples is efficient, but it conflates a computational subset with a geometric neighborhood: random co-occurrence decides which relations are distilled, regardless of their importance under the teacher geometry.

Our key insight is to approximate global geometry by a collection of meaningful local geometries. Rather than treating arbitrary batch subsets as proxies for the corpus, we center a teacher-defined neighborhood at every corpus example. Overlap among these neighborhoods ties the local objectives together, while their composition exposes structure beyond any single neighborhood.

RIPPLE realizes this principle with cached teacher embeddings and a mutual-nearest-neighbor graph. Each graph row defines an immediate local geometry; diffusion expands it into multi-scale targets (Coifman & Lafon, 2006). Sparse candidate selection makes these targets tractable, candidate pooling reuses student encodings, and supervision on overlapping graph rows encourages consistency across anchors. The graph and teacher are discarded at inference.

The analysis supporting this design is deliberately compact. We first show that random mini-batch exposure is independent of teacher relevance. We then bound global teacher-weighted geometry distortion by the alignment error on selected local relations plus the teacher mass left uncovered.

Global geometry $\longrightarrow$ **teacher-defined local geometries** $\longrightarrow$ **multi-scale alignment**

Figure 1: **Global geometry through local alignment.** Conventional relational KD treats batch-local relations as a proxy for corpus geometry. RIPPLE instead constructs teacher-defined local geometries, expands them through diffusion, and realizes them with sparse candidate pools.

Across nine datasets and three teacher–student settings, RIPPLE obtains the best mean compact-student result in all three settings. The gains are strongest and most consistent on semantic textual similarity, whereas pair-classification results are less uniform. Our contributions are:

- We identify random mini-batch geometry as a relevance-agnostic approximation to corpus geometry.
- We introduce RIPPLE, which approximates global geometry with teacher-defined, diffusion-expanded local geometries.
- We demonstrate improved relational preservation and compact-student performance across diverse embedding tasks and teacher–student settings.

2 RELATED WORK

2.1 TEXT EMBEDDING AND RELATIONAL DISTILLATION

Knowledge distillation transfers behavior from a large teacher to a compact student (Hinton et al., 2015). Language-model compression commonly matches output distributions, hidden states, or attention maps (Sanh et al., 2019; Jiao et al., 2020; Wang et al., 2020; Xu et al., 2024), whereas embedding-specific methods target quantities used by similarity-based applications. DistilCSE combines teacher supervision with contrastive learning (Gao et al., 2021a; Xu et al., 2023); EmbedDistill transfers relative query–document geometry (Kim et al., 2023); and recent systems align embeddings, layers, tokens, or heterogeneous representation spaces (Lee et al., 2024; Zhang et al., 2024a; Vujanic & Rueckstiebs, 2025; Truong et al., 2025; Dao et al., 2026; Akram et al., 2026). Relational approaches match pairwise similarities, distances and angles, conditional affinities, graph edges, or topological summaries (Tung & Mori, 2019; Park et al., 2019; Passalis et al., 2021; Liu et al., 2019; Kim et al., 2024). Full Kernel Matrix Transfer uses Nystrom landmarks (Qian et al., 2022), and BINGO groups teacher-similar instances (Xu et al., 2022). Nevertheless, many relational objectives instantiate their relations from the current mini-batch; NRKD likewise mines an in-batch similarity matrix (Gou et al., 2025). This couples computational batching with geometric support selection.

2.2 GRAPH GEOMETRY AND DIFFUSION

Neighborhood graphs recover manifold structure in Laplacian Eigenmaps and Diffusion Maps (Belkin & Niyogi, 2003; Coifman & Lafon, 2006); heat-based constructions provide multi-scale summaries (Sun et al., 2009; Huguet et al., 2023). Geometric KD aligns heat kernels between teacher and student graphs (Yang et al., 2022), while graph-to-MLP methods distill observed graph structure into graphless models (Zhang et al., 2022; Tian et al., 2025). RIPPLE instead induces a graph from cached text embeddings solely to select and diffuse supervision for a standard text encoder; neither graph nor teacher is used at inference.

3 RIPPLE: GLOBAL GEOMETRY THROUGH LOCAL ALIGNMENT

RIPPLE approximates the teacher’s corpus-level geometry through sparse local geometries centered at each example. A cached teacher graph defines immediate neighborhoods, diffusion composes them into broader targets, and pooled candidates realize those targets without full-corpus student scoring. Figure 1 gives the overall view.

3.1 FROM GLOBAL GEOMETRY TO LOCAL ALIGNMENT

Let $\mathcal{D} = \{x_i\}_{i=1}^N$ be an unlabeled corpus, and let t_i and s_i be the unit-normalized teacher and student embeddings. For each anchor i , let p_i^T be a teacher relation distribution over $\mathcal{D} \setminus \{x_i\}$. With teacher and student cosine similarities c_{ij}^T and c_{ij}^S , define the bounded pair distortion

$$\ell_{ij} = \frac{(c_{ij}^S - c_{ij}^T)^2}{4} \in [0, 1]. \quad (1)$$

The quantity

$$\mathcal{E} = \frac{1}{N} \sum_i \sum_{j \neq i} p_i^T(j) \ell_{ij} \quad (2)$$

formalizes the global teacher-weighted geometry we ultimately seek to preserve; it is not an objective that we compute directly.

Conventional relational KD instead observes the examples that happen to share a mini-batch. The following proposition makes the resulting mismatch precise; Appendix A.1 gives the proof.

Proposition 1 (Mini-batch exposure is independent of teacher relevance). *Let $\mathcal{O}_i^{(T)}$ be the examples co-occurring with anchor i over T independent random reshuffles into batches of size B . For any teacher relation distribution p_i^T ,*

$$\mathbb{E} \left[1 - p_i^T(\mathcal{O}_i^{(T)}) \right] = \left(1 - \frac{B-1}{N-1} \right)^T. \quad (3)$$

Thus important and irrelevant relations have the same inclusion probability. This suggests replacing arbitrary batch-local geometry with teacher-defined local geometry. The next proposition explains when a collection of such local objectives is sufficient; its proof is in Appendix A.2.

Proposition 2 (Local-to-global geometry bound). *For any selected relation set C_i with positive teacher mass, define*

$$\epsilon_i = \mathbb{E}_{j \sim p_i^T(\cdot|C_i)} [\ell_{ij}], \quad \delta_i = 1 - p_i^T(C_i), \quad (4)$$

and let $\bar{\epsilon} = N^{-1} \sum_i \epsilon_i$ and $\bar{\delta} = N^{-1} \sum_i \delta_i$. Then

$$\boxed{\mathcal{E} \leq \bar{\epsilon} + \bar{\delta}}. \quad (5)$$

Global preservation therefore reduces to two requirements: accurately align the selected local geometry and cover most teacher-relevant mass. Bounds for teacher-proportional sampling and the concentrated-support comparison are deferred to Appendix A.3 and Appendix A.4.

3.2 TEACHER-DEFINED MULTI-SCALE LOCAL GEOMETRY

We cache the teacher embeddings and form the mutual k -nearest-neighbor set

$$\mathcal{M}_k^T(i) = \text{kNN}_T(i) \cap \{j : i \in \text{kNN}_T(j)\}. \quad (6)$$

Mutuality suppresses one-directional hub edges. Retained edges are weighted by teacher cosine similarity and row-normalized:

$$P_{ij}^T = \frac{\mathbf{1}[j \in \mathcal{M}_k^T(i)] \exp(c_{ij}^T/\tau_i)}{\sum_{\ell \in \mathcal{M}_k^T(i)} \exp(c_{i\ell}^T/\tau_i)}. \quad (7)$$

The row P_i^T defines the immediate local geometry around anchor i . We calibrate τ_i by a common target entropy so that neighborhood sharpness adapts to local density.

One hop preserves immediate local structure; diffusion composes overlapping neighborhoods to expose broader structure. With the lazy operator $\tilde{P}^T = \frac{1}{2}(I + P^T)$, RIPPLE defines

$$g_{i,1}^T = P_i^T, \quad g_{i,r}^T = e_i^\top (\tilde{P}^T)^r \Big|_{\mathcal{D} \setminus \{i\}}, \quad r > 1. \quad (8)$$

These targets provide a multi-scale view of the teacher geometry without requiring a student graph. Graph construction, entropy calibration, and sparse diffusion are detailed in Appendix B.

3.3 SPARSE RELATIONAL MATCHING

Full-corpus scoring would defeat compact training. For every $g_{i,r}^T$, RIPPLE retains high-mass entries, samples additional entries in proportion to teacher mass, and adds hard and uniform negatives. Candidate sets are pooled and deduplicated across anchors, so each candidate text is encoded once while each anchor retains its own target. Let $\Omega_{i,r}$ be the realized domain and $g_{i,r}^S$ the student cosine-softmax on that domain. The relational objective is

$$\mathcal{L}_{\text{rel}} = \frac{1}{|B|} \sum_{i \in B} \sum_{r \in \mathcal{R}} \omega_r D_{\text{KL}}(g_{i,r}^T |_{\Omega_{i,r}} \| g_{i,r}^S |_{\Omega_{i,r}}). \quad (9)$$

The candidate construction also includes an ambient comparison over the shared pool, preventing a restricted high-mass target from hiding off-support alternatives. Exact quotas, scale weights, and temperatures are implementation details given in Appendix B.3.

Anchor-wise objectives are not independent because their local geometries overlap. RIPPLE reuses overlapping pooled candidates to supervise additional rows of the teacher graph. If $\mathcal{J}_B$ is the set of selected row centers, Ω_j^{row} is the available teacher-neighbor support for row j , and $\nu_B(j)$ is its normalized selection weight, then

$$\mathcal{L}_{\text{row}} = \sum_{j \in \mathcal{J}_B} \nu_B(j) D_{\text{KL}}(P_j^T |_{\Omega_j^{\text{row}}} \| \text{softmax}_{\Omega_j^{\text{row}}}(c_j^S / \tau_j)). \quad (10)$$

Rows with fewer than two available teacher-neighbor columns are omitted. The complete objective is

$$\mathcal{L}_{\text{RIPPLE}} = \mathcal{L}_{\text{rel}} + \lambda_{\text{row}} \mathcal{L}_{\text{row}}. \quad (11)$$

The cached graph is used only during distillation; inference uses the student encoder alone.

4 EXPERIMENTS

We first evaluate whether RIPPLE improves compact text embeddings across teacher–student pairs and downstream tasks. Section 5 then tests the mechanisms proposed in Section 3.

4.1 EXPERIMENTAL SETUP

Models and distillation data. We evaluate three teacher→student pairs: Qwen3-Embedding-4B → BERT-base (109M), BGE-M3 → MiniLMv2-H768 (66M), and Qwen3-Embedding-0.6B → MiniLMv2-H384 (22M). The teacher families are described by Zhang et al. (2025) and Chen et al. (2024). Following the TALAS protocol (Dao et al., 2026), the unlabeled distillation corpus contains approximately 15K sentences sampled evenly from Emotion, WiC, and STS-B. Teacher embeddings are cached once; training uses neither task labels nor online teacher inference.

Evaluation. The nine downstream datasets cover classification (Banking77, Tweet, Emotion), pair classification (MRPC, SciTail, WiC), and semantic textual similarity (SICK, STS12, STS-B). We report macro-F1, average precision, and Spearman correlation, respectively. Emotion, WiC, and STS-B are in-domain; the other six datasets are out-of-domain. “Avg.” is the unweighted mean of the nine primary metrics. Checkpoints and pair-classification thresholds are selected on validation data, and test data are evaluated once.

Baselines and implementation. We compare with the teacher, the undistilled student, SimCSE-unsup (Gao et al., 2021b), CDM (Chen et al., 2025), DSKD (Zhang et al., 2024b), Jasper-Stella (Zhang et al., 2024a), DistilCSE (Gao et al., 2021a; Xu et al., 2023), EMO (Truong et al., 2025), and TALAS (Dao et al., 2026). We use one canonical RIPPLE configuration across downstream tasks and vary only the component named in each controlled comparison. Appendices B and C give the method and evaluation details.

4.2 MAIN RESULTS

Table 1 reports all nine datasets for the three teacher–student settings. RIPPLE obtains the best mean compact-student score in each setting: 78.09, 77.99, and 74.91. The corresponding margins over

Table 1: Results across three teacher $\rightarrow$ student settings on nine datasets. Emotion, WiC, and STS-B (marked with $\star$) are in-domain; the remaining datasets are out-of-domain. Bold and underlined values denote the best and second-best results within each setting, excluding the Teacher and Student base reference rows. TALAS and RIPPLE results are reported as mean $\pm$ standard deviation; the remaining baseline results are from TALAS (Dao et al., 2026). For setting (c), the epoch checkpoint is selected on the validation average.

	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Domain Avg		Avg.
	Banking77	Tweet	Emotion $\star$	MRPC	SciTail	WiC $\star$	SICK	STS12	STS-B $\star$	Avg-In	Avg-Out	Avg.
(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)												
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	74.20	84.73	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	42.44	60.33	54.36
SimCSE-unsup	89.21	69.13	53.55	82.86	76.59	71.64	68.19	61.68	70.36	65.18	74.61	71.47
CDM	89.75	70.58	53.11	84.83	75.19	71.06	69.37	68.48	73.92	66.03	76.37	72.92
DSKD	89.66	69.93	54.51	83.13	77.95	71.53	68.65	63.37	71.42	65.82	75.45	72.24
Jasper and Stella	89.71	73.83	65.98	85.33	80.99	70.50	73.96	67.90	75.65	70.71	78.62	75.98
DistillCSE	90.94	70.90	60.80	82.86	78.98	68.63	75.12	72.61	77.08	68.84	78.57	75.32
EMO	90.78	73.54	67.48	84.46	81.57	69.76	75.66	68.83	77.17	71.47	79.14	76.58
TALAS	90.44 $\pm$ 0.13	74.62 $\pm$ 0.17	70.87 $\pm$ 0.52	85.90 $\pm$ 0.10	81.78 $\pm$ 0.29	68.84 $\pm$ 0.15	77.56 $\pm$ 0.10	73.12 $\pm$ 0.30	79.24 $\pm$ 0.10	72.98 $\pm$ 0.19	80.57 $\pm$ 0.01	78.04 $\pm$ 0.07
RIPPLE	91.75 $\pm$ 0.17	73.49 $\pm$ 0.45	70.21 $\pm$ 0.57	83.38 $\pm$ 0.05	83.47 $\pm$ 0.18	66.58 $\pm$ 0.06	77.21 $\pm$ 0.13	75.48 $\pm$ 0.19	81.23 $\pm$ 0.09	72.67 $\pm$ 0.14	80.80 $\pm$ 0.04	78.09 $\pm$ 0.04
(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)												
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	71.63	83.83	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	44.24	62.41	56.35
SimCSE-unsup	87.89	71.45	55.65	83.87	76.32	67.96	66.93	59.66	63.97	62.53	74.35	70.41
CDM	89.15	70.48	58.71	84.16	75.91	70.00	68.14	62.74	67.78	65.50	75.10	71.90
DSKD	89.51	70.53	57.27	84.39	75.36	69.96	68.81	66.78	70.05	65.76	75.90	72.52
Jasper and Stella	88.38	74.13	62.81	85.98	80.33	68.38	74.35	66.80	74.55	68.58	78.33	75.08
DistillCSE	90.50	72.70	59.88	84.39	78.48	69.49	73.97	70.00	73.73	67.70	78.34	74.79
EMO	90.19	74.26	61.48	85.31	80.82	68.30	75.41	67.06	76.35	68.71	78.84	75.46
TALAS	89.72 $\pm$ 0.19	75.38 $\pm$ 0.11	65.83 $\pm$ 0.58	87.58 $\pm$ 0.02	82.90 $\pm$ 0.12	66.96 $\pm$ 0.07	75.82 $\pm$ 0.11	68.11 $\pm$ 0.20	77.51 $\pm$ 0.18	70.10 $\pm$ 0.13	79.92 $\pm$ 0.05	76.64 $\pm$ 0.04
RIPPLE	90.58 $\pm$ 0.12	73.97 $\pm$ 0.30	71.06 $\pm$ 0.98	85.22 $\pm$ 0.09	83.92 $\pm$ 0.14	62.87 $\pm$ 0.15	78.81 $\pm$ 0.09	73.99 $\pm$ 0.20	81.46 $\pm$ 0.22	71.80 $\pm$ 0.31	81.08 $\pm$ 0.07	77.99 $\pm$ 0.13
(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)												
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	72.66	82.61	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	40.91	58.45	52.60
SimCSE-unsup	86.67	69.56	54.00	82.29	73.76	65.71	66.04	59.11	62.08	60.60	72.91	68.80
CDM	86.07	70.32	53.04	81.97	72.97	68.34	65.97	61.12	65.75	62.38	73.07	69.51
DSKD	85.66	69.86	52.25	82.17	73.52	68.58	66.54	62.52	66.70	62.51	73.38	69.76
Jasper and Stella	85.94	71.25	57.99	83.36	76.17	68.25	70.57	63.81	70.11	65.45	75.18	71.94
DistillCSE	88.03	69.87	54.24	83.20	77.66	67.88	70.09	68.19	71.53	64.55	76.17	72.30
EMO	87.03	71.67	59.63	83.45	78.17	68.89	70.81	65.78	71.42	66.65	76.15	72.98
TALAS	84.16 $\pm$ 0.15	72.82 $\pm$ 0.11	61.13 $\pm$ 0.52	84.85 $\pm$ 0.10	80.91 $\pm$ 0.27	66.32 $\pm$ 0.07	70.72 $\pm$ 0.15	67.80 $\pm$ 0.08	72.39 $\pm$ 0.31	66.61 $\pm$ 0.24	76.88 $\pm$ 0.07	73.46 $\pm$ 0.10
RIPPLE	87.50 $\pm$ 0.24	70.50 $\pm$ 0.29	66.40 $\pm$ 0.40	81.68 $\pm$ 0.12	80.73 $\pm$ 0.05	65.34 $\pm$ 0.10	73.69 $\pm$ 0.11	72.08 $\pm$ 0.09	76.29 $\pm$ 0.08	69.34 $\pm$ 0.13	77.70 $\pm$ 0.11	74.91 $\pm$ 0.08

the strongest baseline are 0.05, 1.35, and 1.45 points; the first is small relative to run variation. The most consistent gains occur on semantic textual similarity. RIPPLE gives the best compact-student STS-B result in all three settings and leads all three STS datasets for both MiniLMv2 students. Pair classification is less uniform, including lower WiC scores. The table therefore supports RIPPLE’s overall competitiveness, particularly on similarity-sensitive tasks, but does not by itself identify which mechanism produces the gains.

5 ANALYSIS

The analysis follows the method’s causal chain: whether teacher-defined locality improves the support, whether diffusion transfers structure beyond one hop, and how much sparse realization costs. We interpret downstream scores together with full-corpus relational diagnostics rather than using task performance alone as evidence for a mechanism.

5.1 DOES TEACHER-DEFINED LOCALITY MATTER?

Our primary matched-compute comparison is

$$\text{batch-local support} \rightarrow \text{random corpus support} \rightarrow \text{mutual-}k\text{NN support.} \quad (12)$$

The first transition asks whether broader corpus exposure alone is sufficient; the second asks whether the teacher’s local geometry matters when the number of anchor-column scores is held fixed. We report downstream average, full-corpus JS divergence, and retained teacher mass. All variants share the student initialization, optimizer, schedule, seeds, and scored-pair budget; Appendix D.1 specifies the controls.

5.2 DOES DIFFUSION RECOVER BROADER GEOMETRY?

We compare one-hop RIPPLE with the same model augmented by diffusion targets. The main evidence is teacher-student error on held-out broader-rank relations, full-corpus relational divergence, and downstream performance. Improvement outside the immediate mutual- k NN support would show that diffusion transfers broader geometry rather than merely reweighting direct neighbors.

Matched-compute locality analysis.

Batch-local $\rightarrow$ random corpus $\rightarrow$ mutual- k NN.
 Downstream Avg. $\uparrow$ Full-corpus JS $\downarrow$ Retained mass $\uparrow$.

Figure 2: **Does teacher-defined locality matter?** Random corpus support controls for broader exposure, while mutual- k NN support tests whether teacher relevance improves geometry preservation at matched relational compute. Bars report mean $\pm$ standard deviation over seeds.

Accuracy-efficiency analysis.

Candidate budget vs. retained mass, downstream quality,
 full-corpus JS, step time, and peak memory.

Figure 3: **Accuracy-efficiency trade-off.** Candidate budgets are evaluated through target coverage, relational preservation, downstream quality, and computational cost.

Cross-anchor row supervision appears as an additional row in the component ablation rather than a separate main-paper study. The fuller controls—including injection-only supervision, graph-width sweeps, fixed rank bands, row-overlap disagreement, and a degree-preserving random graph—are reported in Appendix D.

5.3 ACCURACY-EFFICIENCY TRADE-OFF

We vary the candidate budget and jointly report retained teacher mass, downstream average, full-corpus JS divergence, step time, and peak memory. This reveals whether the local targets saturate before the cost approaches full-corpus scoring. The same sweep also tracks the two terms in Proposition 2: selected-support distortion $\bar{\epsilon}$ and missed teacher mass $\bar{\delta}$.

Until these mechanism diagnostics are populated, Table 1 supports effectiveness but not a causal attribution to locality, diffusion, or row supervision.

6 CONCLUSION

RIPPLE approximates corpus-level teacher geometry through teacher-defined local geometries, diffusion-expanded targets, and sparse pooled matching. Its compact theoretical justification isolates the weakness of random batch exposure and separates global distortion into local alignment error and uncovered teacher mass. Across three teacher-student settings, RIPPLE is strongest and most consistent on semantic textual similarity and adds no inference-time machinery: deployment uses only the distilled encoder.

The method remains limited by the teacher and corpus used to construct its graph. Teacher errors, hubs, domain bias, and heavy-tailed relation distributions can weaken the value of local support; row supervision is also limited by overlap in the shared candidate pool. Exact neighbor construction is quadratic before sparsification, although approximate search can replace it at larger scale. Finally, the geometry bound does not guarantee downstream improvement, consistent with the less uniform pair-classification results.

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A THEORY

A.1 PROOF OF MINI-BATCH EXPOSURE

Proof of Proposition 1. For a fixed anchor i and relation $j \neq i$, a uniformly shuffled batch of size B includes j with probability $(B - 1)/(N - 1)$. Independence across reshuffles therefore gives

$$\Pr[j \notin \mathcal{O}_i^{(T)}] = \left(1 - \frac{B - 1}{N - 1}\right)^T. \quad (13)$$

Weighting this event by $p_i^T(j)$ and summing over j proves Eq. (3), because $\sum_j p_i^T(j) = 1$. $\square$

A.2 PROOF OF THE LOCAL-TO-GLOBAL BOUND

Proof of Proposition 2. Splitting the distortion for anchor i over selected and unselected relations gives

$$\sum_{j \neq i} p_i^T(j) \ell_{ij} = p_i^T(C_i) \epsilon_i + \sum_{j \notin C_i} p_i^T(j) \ell_{ij} \quad (14)$$

$$\leq \epsilon_i + \delta_i, \quad (15)$$

because $p_i^T(C_i) \leq 1$ and $0 \leq \ell_{ij} \leq 1$. Averaging over anchors proves $\mathcal{E} \leq \bar{\epsilon} + \bar{\delta}$. $\square$

A.3 TEACHER-PROPORTIONAL MISSED-MASS BOUND

Let $C_i^{(T)}$ be the union of relations selected for anchor i through epoch T , and let $\bar{\delta}_T = N^{-1} \sum_i [1 - p_i^T(C_i^{(T)})]$. Suppose RIPPLE makes at least m teacher-proportional draws per anchor and epoch. A relation retained deterministically has zero miss probability. For any remaining relation j , proportional sampling without replacement selects j at each draw with conditional probability at least

Component	Configuration
Teacher graph	Mutual k NN, $k = 200$; target perplexity 30
Diffusion	Scales $\{1, 2, 4\}$; truncation tolerance 0.01
Candidates	66 per anchor: 14 target, 26 hard, 26 random
Row supervision	4 walks of length 4; start epoch 2; $\lambda_{\text{row}} = 0.5$
Ambient profile	Teacher/student temperature 0.10

Table 2: Canonical RIPPLE method configuration. Controlled comparisons vary only the named component.

$p_i^T(j)$, because removing other entries can only decrease the remaining normalizer. Its probability of being missed by at least mT draws is therefore at most $(1 - p_i^T(j))^{mT}$. Consequently,

$$\mathbb{E}[\bar{\delta}_T] \leq \frac{1}{N} \sum_i \sum_{j \neq i} p_i^T(j) (1 - p_i^T(j))^{mT}. \quad (16)$$

Deterministic high-mass retention and additional candidates can only reduce missed mass.

Combining Eq. (16) with Proposition 2 yields the finite-epoch statement omitted from the main paper: the final global distortion is bounded by its selected-support error plus the residual teacher mass implied by the sampling process.

A.4 CONCENTRATED-SUPPORT COMPARISON

When B is small relative to N , Proposition 1 requires $\Theta(N \log(1/\varepsilon))$ batch relation opportunities per anchor to reduce expected missed mass below ε . If p_i^T is uniform on h relevant relations, Eq. (16) reduces to

$$\mathbb{E}[\delta_{i,T}] \leq (1 - 1/h)^{mT}, \quad (17)$$

which requires $\Theta(h \log(1/\varepsilon))$ teacher-proportional draws. Thus matched relational compute gives a factor- $\Theta(N/h)$ separation in the concentrated-support regime. This comparison is not distribution-free: real teacher targets can be heavy-tailed, in which case the empirical retained-mass curve determines the advantage.

B METHOD DETAILS

B.1 CANONICAL METHOD CONFIGURATION

Teacher embeddings, graph neighborhoods, and diffusion targets are computed once before training. The teacher is not loaded during student optimization, and all downstream tasks use the same method configuration.

B.2 GRAPH CONSTRUCTION AND ENTROPY CALIBRATION

We compute cosine nearest neighbors from unit-normalized cached teacher embeddings, retain reciprocal edges, and store the resulting graph sparsely. A node with no reciprocal edge falls back to its ordinary nearest-neighbor row so that Eq. (7) remains defined.

For each non-degenerate row, bisection selects τ_i such that

$$H(P_i^T) = \log \rho, \quad (18)$$

where the target perplexity is $\rho = 30$. Rows whose degree is below ρ use their largest feasible entropy; degree-one and constant-score rows use the uniform distribution on their available support. The calibrated temperatures are stored with the graph and reused for one-hop student readouts and row supervision.

B.3 SPARSE DIFFUSION AND CANDIDATE REALIZATION

The lazy transition $\tilde{P}^T = \frac{1}{2}(I + P^T)$ prevents periodic oscillation. After each sparse multiplication, we retain the smallest support carrying at least 99% of the row mass and renormalize. If discarded

mass is at most η , the truncated distribution differs from the original by total variation η and has reverse KL $-\log(1 - \eta)$; the canonical tolerance is $\eta = 0.01$. Student temperatures at scale r are derived as $\sqrt{r} \tau_i$, matching the increasing spread of the diffusion target.

For each anchor and epoch, the sampler keeps the two highest-mass entries, fills the remainder of its target quota without replacement in proportion to teacher mass, and adds hard and random negatives. Across an anchor batch, candidates are pooled and deduplicated before student encoding. The canonical relational scale set is $\mathcal{R} = \{0, 1, 2, 4\}$: $r = 0$ is an ambient teacher/student cosine-softmax over the shared pool at temperature 0.10, $r = 1$ is the direct transition row, and $r > 1$ denotes diffusion. Scale weights follow the fixed rule

$$\omega_r \propto \frac{1}{\max(1, r)}, \quad (19)$$

normalized once over $\mathcal{R}$.

B.4 CROSS-ANCHOR ROW SUPERVISION

From each anchor, RIPPLE samples four non-backtracking walks of length four using P^T . The walks select source rows; they are not prediction trajectories. Visited nodes replace random candidates when necessary. For a selected node j , the realized row support is

$$\Omega_j^{\text{row}} = \mathcal{P}_B \cap \mathcal{M}_k^T(j), \quad (20)$$

where $\mathcal{P}_B$ is the shared candidate pool. The loss in Eq. (10) matches the complete available transition row rather than a sampled successor. Duplicate visits are combined through $\nu_B(j)$, and rows with fewer than two available columns are skipped.

B.5 COMPLEXITY AND APPROXIMATE NEIGHBOR SEARCH

Exact construction from N embeddings of dimension d requires $O(N^2d)$ similarity work before sparsification. The stored mutual-neighbor graph uses $O(Nk)$ edges, and sparse diffusion operates on this cached graph subject to the mass-based row cap. During training, student computation depends on the deduplicated batch candidate pool rather than all N corpus items; the canonical pre-deduplication budget is 66 candidates per anchor. At inference, both graph storage and graph operations disappear.

For larger corpora, approximate nearest-neighbor search can replace the exact all-pairs stage. The remaining pipeline only requires neighbor indices and teacher similarities, so mutuality filtering, entropy calibration, diffusion, and candidate sampling are unchanged. Approximation quality should be audited through reciprocal-edge recall and retained teacher mass because missed high-relevance neighbors directly affect the coverage term in Proposition 2.

C EXPERIMENTAL DETAILS

C.1 MODELS AND DISTILLATION CORPUS

We evaluate Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M), BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M), and Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M). The unlabeled corpus contains approximately 15,000 sentences, sampled evenly from the training splits of Emotion, WiC, and STS-B. Corpus texts are deduplicated before graph construction. Teacher embeddings are unit-normalized, cached once, and reused by every method variant that requires teacher representations.

C.2 OPTIMIZATION AND REPRODUCIBILITY

Stochastic candidate and walk streams are seeded independently by the run seed, epoch, anchor index, and stream identifier. Results for RIPPLE and TALAS are reported as mean $\pm$ standard deviation across three seeds. Comparisons of step time and peak memory use the same device, batch, mixed-precision policy, software stack, and measurement procedure for every variant; the exact accelerator and library versions should be included with the released run manifest.

Component	Configuration
Optimizer	AdamW
Batch size and duration	32 anchors; 5 epochs
Learning-rate schedule	2×10^{-5} ; 6% warm-up; cosine decay
Minimum learning rate	3×10^{-6}
Random seeds	42, 43, 44
Checkpoint selection	Validation average

Table 3: Training and selection protocol used for RIPPLE.

C.3 DOWNSTREAM PROTOCOLS

The nine evaluation datasets are grouped into classification (Banking77, Tweet, Emotion), pair classification (MRPC, SciTail, WiC), and semantic textual similarity (SICK, STS12, STS-B). For classification, a logistic-regression probe is fit on frozen embeddings and evaluated by macro-F1. Pair classification uses cosine similarity and is evaluated by average precision; any decision threshold used for auxiliary classification metrics is selected on validation data. STS uses the Spearman correlation between cosine similarity and human similarity scores. Emotion, WiC, and STS-B are treated as in-domain because their training data contribute unlabeled corpus sentences; the other six datasets are out-of-domain.

Checkpoints are selected using validation data, after which the corresponding test split is evaluated once. The primary aggregate is the unweighted mean of the nine task metrics. No downstream labels enter distillation or teacher-graph construction.

C.4 BASELINES

The comparison includes the frozen teacher and undistilled student references, SimCSE-unsup, CDM, DSKD, Jasper–Stella, DistillCSE, EMO, and TALAS. Published baseline results are taken from the shared TALAS protocol where indicated in Table 1; TALAS and RIPPLE are summarized over three runs. Controlled RIPPLE comparisons use the same initialization, optimizer, schedule, and evaluation code and vary only the named component.

D ADDITIONAL ANALYSIS PROTOCOLS

D.1 MATCHED-COMPUTE COMPONENT ABLATION

All variants use the same student initialization, optimizer, epochs, seeds, and scored anchor–column budget. Batch-local KD normalizes over co-occurring examples; random corpus support draws a uniformly sampled support of matched size; one-step RIPPLE uses only direct mutual-neighbor rows; diffusion adds broader targets; and full RIPPLE adds cross-anchor row supervision. An injection-only control uses the same graph traversals and candidate replacement with $\lambda_{\text{row}} = 0$. A degree-preserving random-graph control rewires adjacency before applying the same targets, sampler, and loss construction.

The primary locality comparison reports downstream average, full-corpus teacher–student JS divergence, and retained teacher mass. The extended table additionally reports teacher-neighbor recall and student off-support mass. This separation prevents low restricted-support KL from being interpreted as global agreement.

D.2 DIFFUSION AND GRAPH WIDTH

One-hop and diffusion variants are evaluated on teacher–student error in fixed teacher-rank bands and on held-out diffusion profiles at scales broader than those directly supervised. Graph-width sweeps keep candidate budget and optimization fixed. Because increasing k trivially changes available support, every width is evaluated on the same fixed rank bands as well as on full-corpus divergence.

The degree-preserving random-graph control tests whether gains follow from teacher geometry rather than generic graph regularization. For each rewired graph, degree sequence, target scales, candidate quota, and update count are held fixed.

D.3 CANDIDATE BUDGET AND GRADIENT FIDELITY

Candidate-budget sweeps report retained teacher mass, selected-support distortion $\bar{\epsilon}$, missed mass $\bar{\delta}$, their sum, full-corpus distortion $\mathcal{E}$, full-corpus JS, downstream average, step time, and peak memory. For small corpus slices where full scoring is feasible, gradient fidelity is measured by the cosine similarity between the sparse-candidate gradient and the full-corpus relational gradient under the same anchors and student checkpoint.

D.4 CROSS-ANCHOR ROW DIAGNOSTICS

Row-supervision comparisons report distinct selected rows, row hit rate, the effective number of available columns, transition-row KL, and disagreement on graph rows shared by multiple anchors. The injection-only control separates the row loss from changes in candidate composition. All diagnostics use the same anchor set and pooled candidates across compared variants.